



Unifying the Gutenberg-Richter Law with Probabilistic Catalog Completeness

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RISKS-X

风险分析预测与管控研究院

1. Motivation

Core Challenges in FMD Modeling

GR Law Limitations:

- Empirical FMD deviates from GR at low magnitudes ($m < m_c$)
- Fixed m_c approach oversimplifies gradual detection probability
- Deviations from the GR law are more likely observational artifacts than intrinsic physical effects.

Critical Need:

- Unified framework coupling detection probability & GR distribution
- Resolve asymmetry, nonlinearity, parameter interdependence

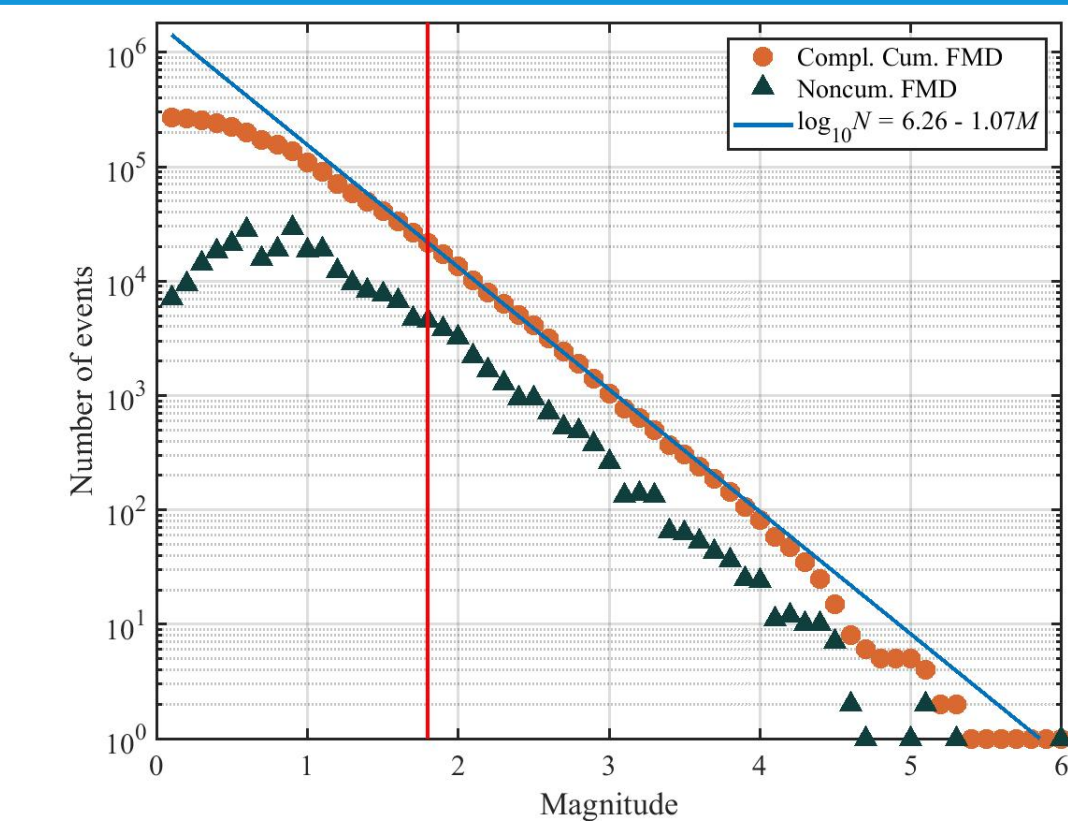


Figure 1. Complementary cumulative distribution (CCFMD) and density distribution (FMD) alongside the Gutenberg-Richter (GR) law, fitted using simulated earthquakes with magnitudes larger than 1.8.

2. Models:

An augmented Gutenberg-Richter (GR) law tailored to handle incompleteness, expressed as:

$$\text{CCFMD}(m) = 10^{a-b \cdot G(m)}$$

in which,

$$G(m) = m_c + \sigma_c \cdot \text{gReLU}(x)$$

where $x = (m - m_c)/\sigma_c$

- m_c : replaces the completeness magnitude as a transition magnitude around which earthquake detection change rapidly.
- σ_c : quantifies the magnitude range over which this transition occurs.

3. gReLU

- SSReLU.**

$$\text{SSReLU}(x) = \ln(1 + e^x)$$

- BSReLU.**

$$\text{BSReLU}(x) = \left(x + \frac{m_c}{\sigma_c} \right) \Phi \left(\frac{\ln[(x\sigma_c + m_c)/m_c]}{\sigma_c} + \frac{\sigma_c}{2} \right) - \frac{m_c}{\sigma_c} \Phi \left(\frac{\ln[(x\sigma_c + m_c)/m_c]}{\sigma_c} - \frac{\sigma_c}{2} \right)$$

- COReLU.**

$$\text{COReLU}(x) = \begin{cases} e^{x-1} & x < 1 \\ x & x \geq 1 \end{cases}$$

- AEReLU.**

$$\text{AEReLU}(x) = \begin{cases} \frac{e^x}{1+\beta} & x < 0 \\ x + \frac{e^{-\beta x}}{1+\beta} & x \geq 0 \end{cases}$$

β : controlling the degree of asymmetry

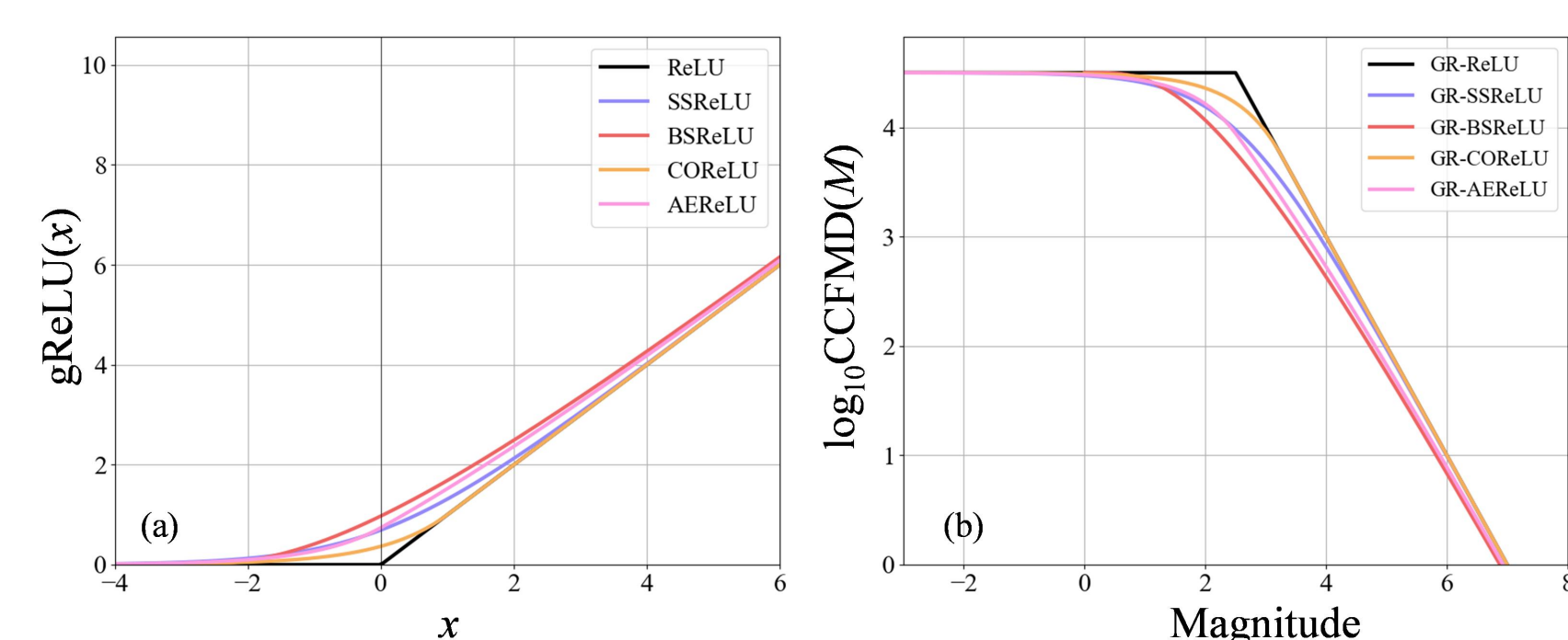


Figure 2. Plots of (a) the four gReLU(x) functions introduced in the text and (b) the corresponding augmented complementary cumulative frequency-magnitude distributions (CCFMDs). For comparison, the figure also includes the Rectified Linear Unit, ReLU(x), depicted with a black solid line, which is defined as ReLU(x) = 0 for $x < 0$ and ReLU(x) = x for $x \geq 0$, along with the corresponding CCFMD recovering the pure Gutenberg-Richter law. The relationship between x in (a) and the magnitude m in (b) is given by $x = (m - m_c)/\sigma_c$. The parameters are set to $a = 7.0$, $b = 1.0$, $m_c = 2.5$, $\sigma_c = 0.75$, and $\beta = 0.35$, which are values that are informed by empirical data.

4. Earthquake catalog

Synthetic Catalogs

- Generation:

Sampled from 4 gReLU(x) - augmented GR models

PDF bounds: $m_{\min} = 0$, $m_{\max} = 8.0$

Fixed params: $b^{\text{true}} = 1.0$, $m_c^{\text{true}} = 2.5$, $\sigma_c^{\text{true}} = 0.75$, and $\beta^{\text{true}} = 0.35$

- Design:

16 datasets(4 model * 4 catalog sizes 10^k , $k = 3, 4, 5, 6$)

200 realizations per dataset

Observed Catalogs

- 6 Seismic Regions:

China: Beijing-Tianjin-Hebei, the southeastern coast, Sichuan-Yunnan, and northern Xinjiang

Global: California, New Zealand

- Key Features:

Period: 2009 - 2023/2024

Boundaries: $m_c \approx 1.8$ contours(BMC method) or station-density-based criteria

Sources: CENC, ANSS ComCat, GeoNet

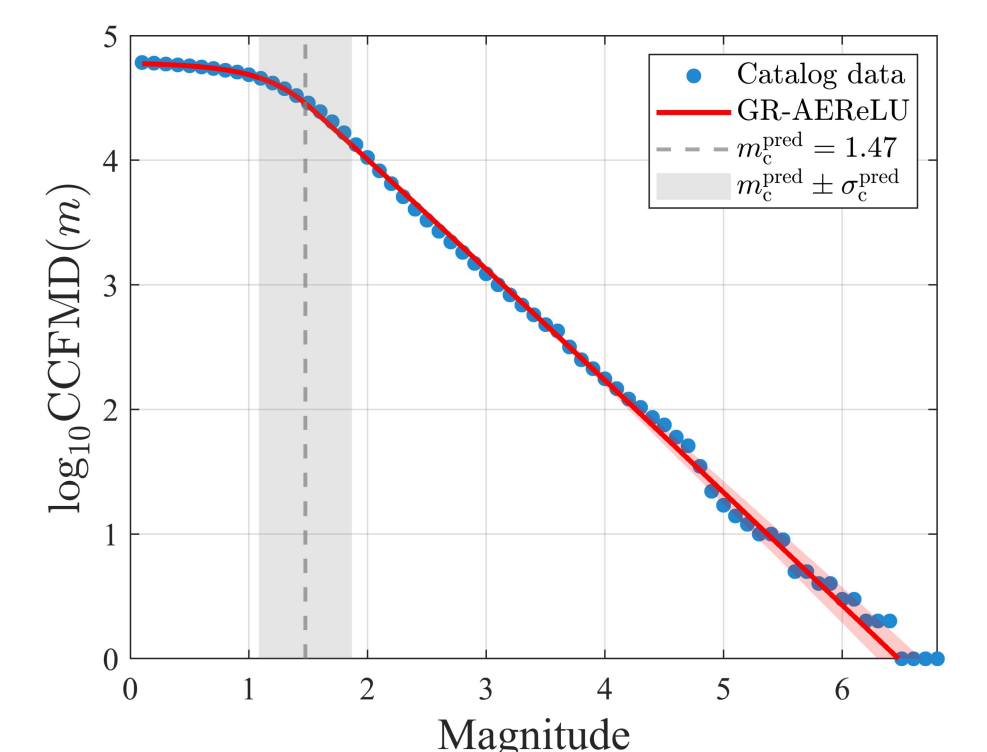


Figure 3. CCFMD of the empirical earthquake catalogs from northern Xinjiang, along with the fitted CCFMD(m) derived using the augmented Gutenberg-Richter law with the AEReLU(x) function. The red shading represents the estimated uncertainty, obtained from 200 bootstrapping iterations.

5. Results

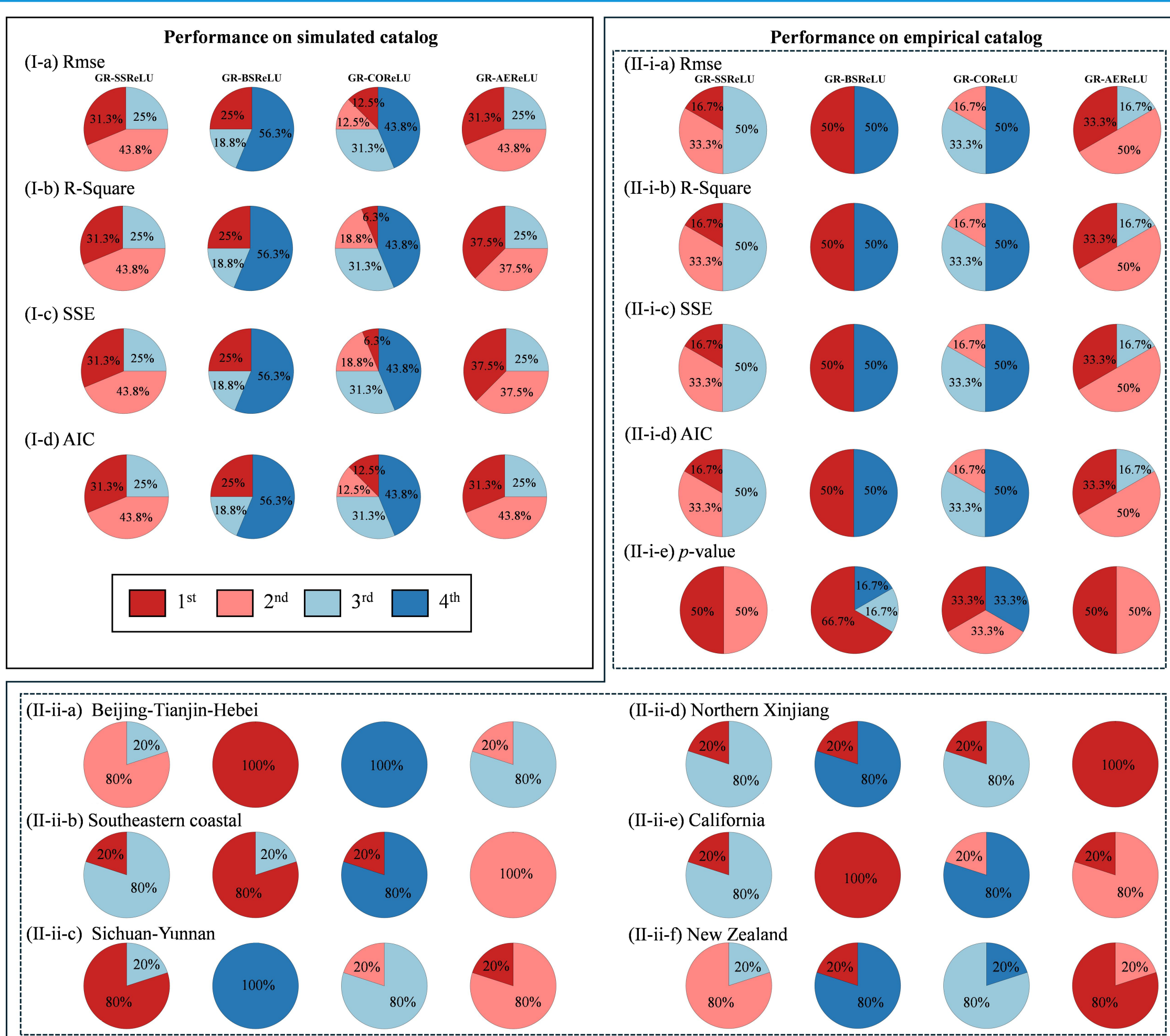


Figure 4. Proportions of times each model (GR-SSReLU, GR-BSReLU, GR-COReLU, and GR-AEReLU) ranked as best (1st, dark red), good (2nd, light red), poor (3rd, light blue), or worst (4th, dark blue) across different evaluation settings. Panels (I) summarize model rankings based on four metrics (Rmse, R-Square, SSE, and AIC) for simulated catalogs and five metrics (including p -value) for empirical catalogs. Panels (II-i) and (II-ii) present alternative views of the same data from Table S5: (II-i) shows, for each metric, how often each model ranks 1st to 4th across the six study regions, while (II-ii) shows, for each region, how often each model ranks 1st to 4th across the five evaluation metrics.

6. Magnitude-dependent catalog completeness probability

The catalog completeness probability $P(m)$: the probability that events of magnitude m or larger are recorded in a given region.

$$P(m) = \frac{10^{a-b \cdot G(m)}}{10^{a-b \cdot m}} = 10^{b[m-G(m)]}$$

Substituting the expression for $G(m)$, we obtain:

$$P(m) = 10^{b\sigma_c \left[\frac{m-m_c}{\sigma_c} - \text{gReLU}\left(\frac{m-m_c}{\sigma_c}\right) \right]}$$

Thus, by substituting the gReLU(x) expressions and the corresponding parameter estimates into Equation, one can compute the completeness probability for any given magnitude m .

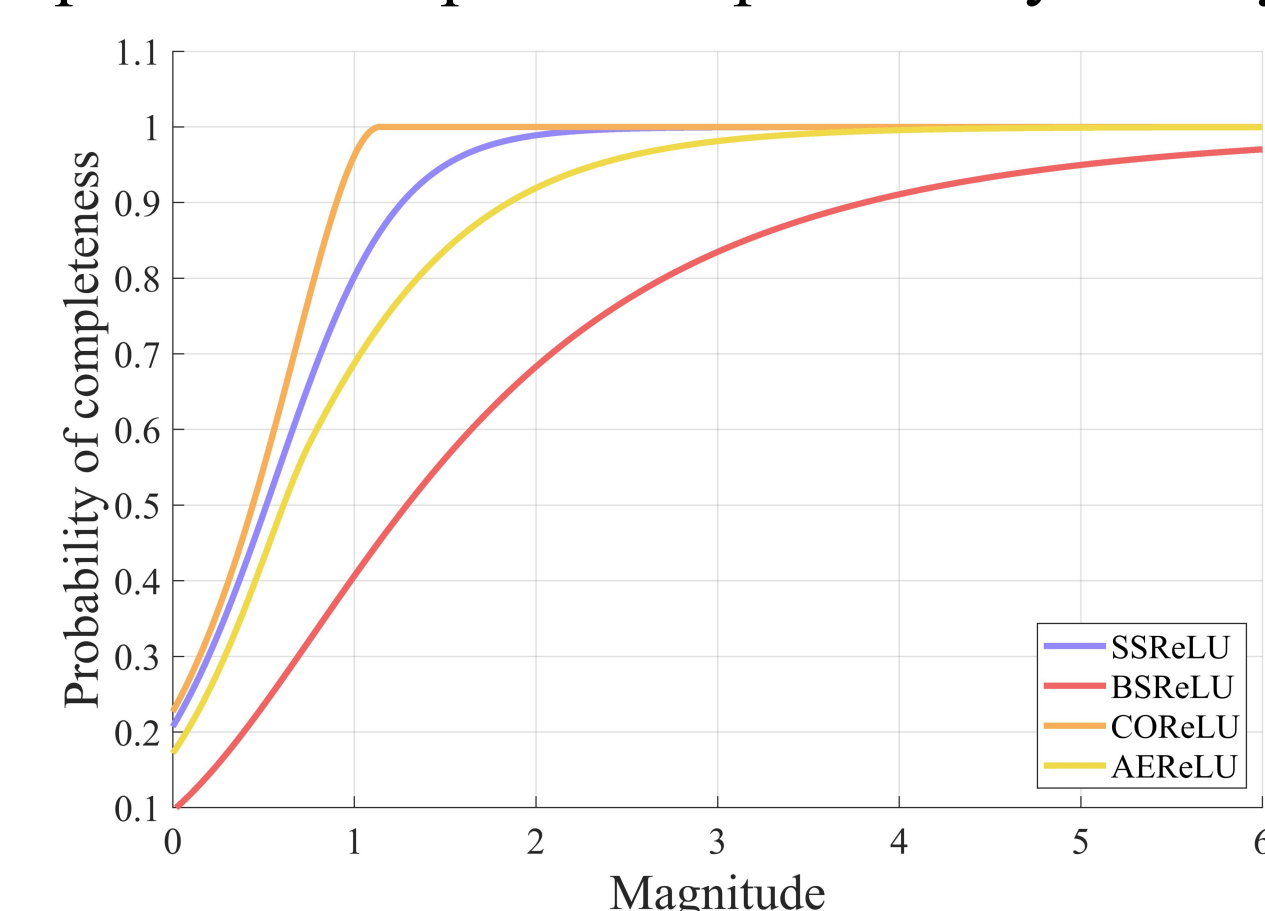


Figure 6. Probability of completeness $P(m)$ for the Beijing-Tianjin-Hebei region.

7. Key Points:

(1) We propose an augmented modeling framework that unifies the Gutenberg-Richter law with a probabilistic treatment of catalog incompleteness.

(2) We evaluate four variants of the proposed Gutenberg-Richter model using both synthetic and empirical catalogs across multiple performance metrics.

(3) The GR-AEReLU, allowing for an asymmetric crossover with two distinct exponential decays, consistently outperforms other variants and offers valuable seismological insights.

(4) The earthquake b -value varies systematically by region and tectonic setting, not the universal constant of 1.0 assumed in standard seismology.



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Code & Data: https://github.com/lijiawei098/GR-gReLU

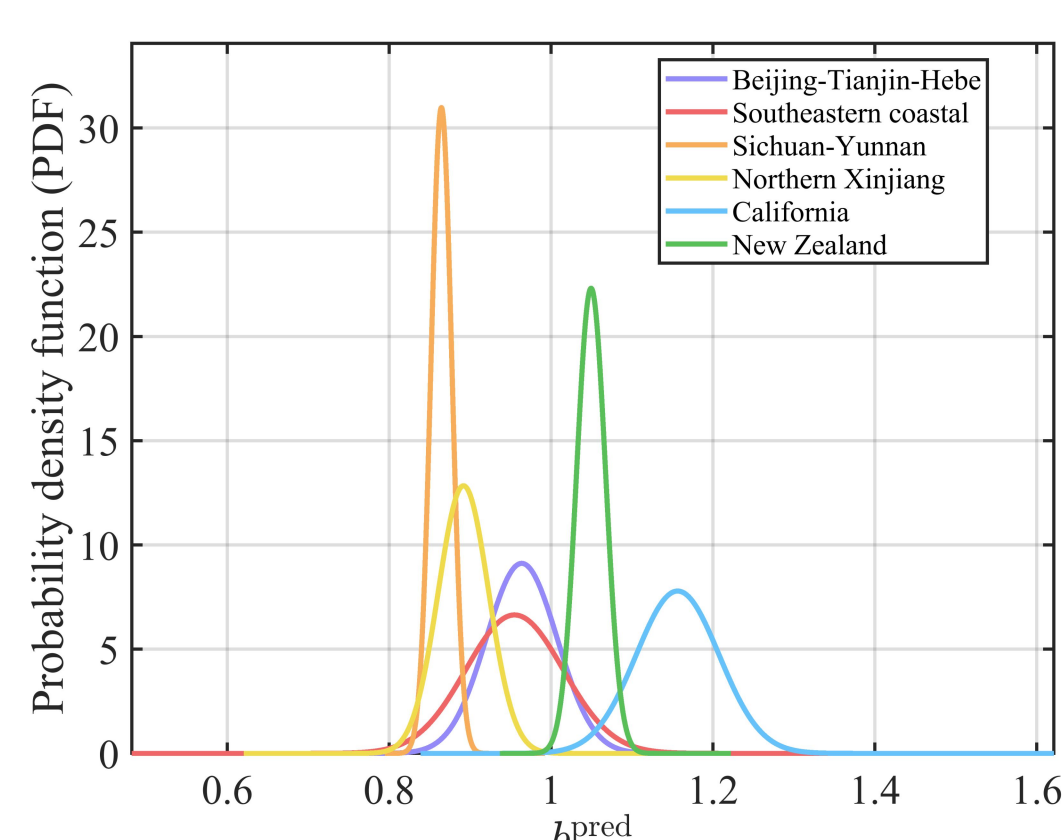


Figure 5. Weighted average of the b^{pred} ,value probability density functions (PDFs) from the four augmented Gutenberg-Richter models for the six study regions.

$$\ln[\text{PDF}_{\text{weight}}(b^{\text{pred}})] = \frac{1}{4} \sum_{i=1}^4 \omega_i \ln[\text{PDF}_i(b^{\text{pred}})]$$

$$\omega_i = \frac{\exp(-\frac{1}{2}\Delta\text{AIC}_i)}{\sum_{j=1}^4 \exp(-\frac{1}{2}\Delta\text{AIC}_j)}$$

with $\Delta\text{AIC}_i = [\text{AIC}_i - \min(\text{AIC})]/[\max(\text{AIC}) - \min(\text{AIC})]$